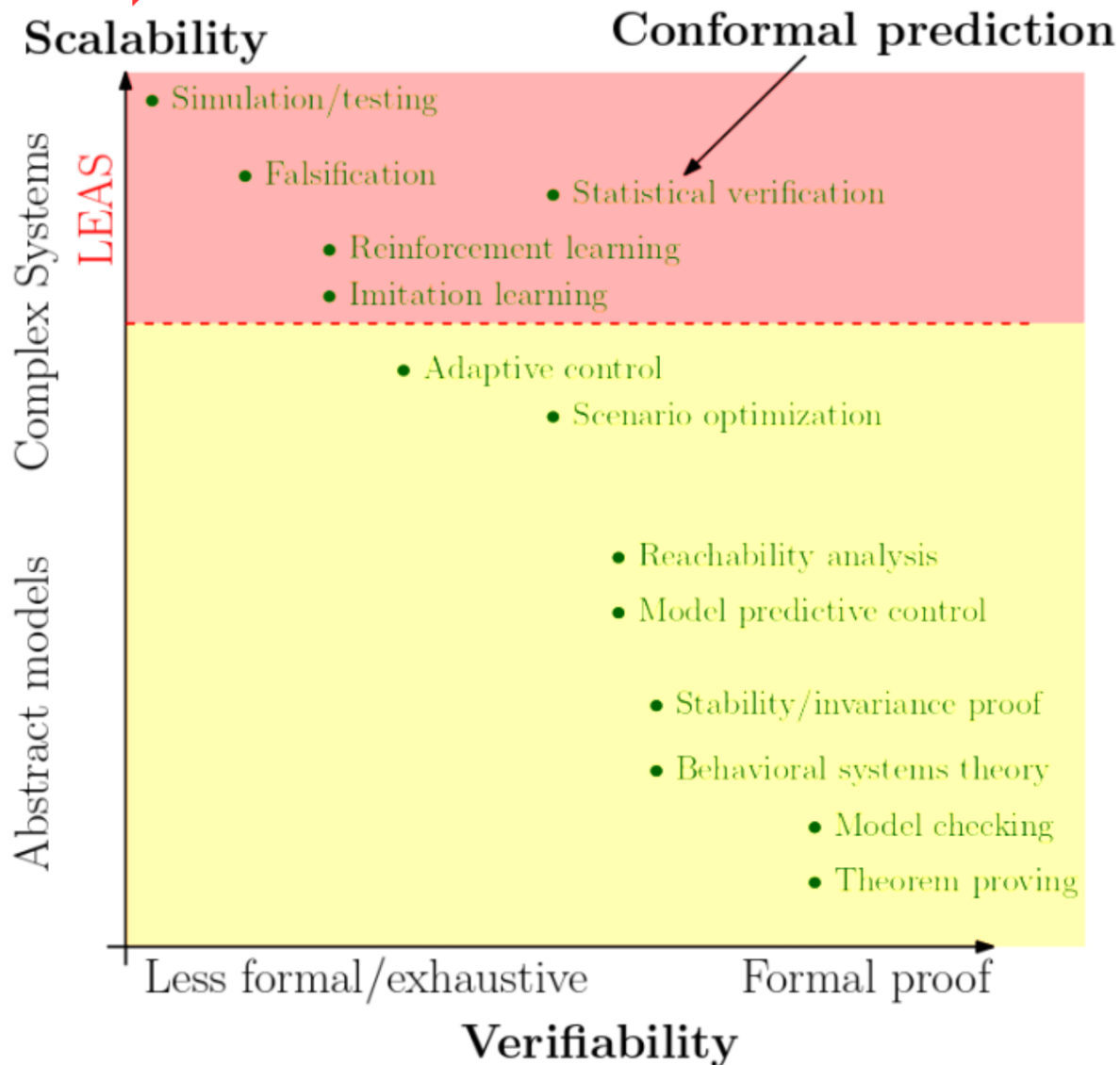


Conformal Prediction

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MEASURE THEORY NOTES

let Ω BE AN ABSTRACT PROBABILITY SPACE

call $(\Omega, \mathcal{F}, \mathbb{P})$ A PROBABILITY SPACE,

WHERE $\mathcal{F}$ IS A σ -ALGEBRA OF

SUBSETS OF Ω AND $\mathbb{P}: \mathcal{F} \rightarrow [0, 1]$

IS A PROBABILITY MEASURE DEFINED ON $\mathcal{F}$

σ - ALGEBRA AXIOMS

FOR AN EVENT SPACE $\mathcal{F}$ (SET OF SETS OF OUTCOMES IN Ω) TO REPRESENT A σ -ALGEBRA, IT MUST SATISFY:

- 1) CONTAINS BOTH EMPTY SET $\emptyset$ AND Ω
- 2) CLOSED UNDER COMPLEMENT: $\forall \alpha \in \mathcal{F}, \alpha^c \in \mathcal{F}$
- 3) CLOSED UNDER (COUNTABLE) UNION: $\forall \alpha, \beta \in \mathcal{F}, \alpha \cup \beta \in \mathcal{F}$
 - (IMPLIES): CLOSED UNDER (COUNTABLE) INTERSECTION: $\alpha \cap \beta = (\alpha^c \cup \beta^c)^c$

P - MEASURE AXIOMS

- 1) $P(\alpha) \in [0, 1] \quad \forall \alpha \in \mathcal{F}$
- 2) $P(\emptyset) = 0$
- 3) $P(\Omega) = 1$
- 4) IF $\alpha \cap \beta = \emptyset$ THEN $P(\alpha \cup \beta) = P(\alpha) + P(\beta)$
(IN THIS CASE, α AND β ARE DISJOINT)

SET PREDICTOR (WITH $m \in \mathbb{N}$)

GIVEN DATA SET $\mathcal{S} = \{w_i\}_{i=1}^m \subset \Omega^m$

A SET PREDICTOR $(T^L: \Omega^m \mapsto 2^\Omega)$ POWER SET OF Ω
IS A SET-VALUED MAP SENDING AVAILABLE DATA
TO A COLLECTION OF REALISATIONS THAT ARE
USED TO INFORM THE PREDICTION OF AN
UNKNOWN OUTCOME. TEST SAMPLE $:= w$

$$T^L(\mathcal{S}) \subseteq \Omega \quad (\text{OUTPUT IS A SUBSET OF } \Omega)$$

AS OPPOSED TO A SINGLE-VALUED FUNCTION

$$f: \Omega^m \mapsto \Omega \quad \left(T^L(\mathcal{S}) \overset{T^L(\mathcal{S}) \text{ GENERALISES } f(\mathcal{S})}{=} \{f(\mathcal{S})\} \right)$$

STANDING ASSUMPTION 1

FOR ANY $m \in \mathbb{N}$, THE SUBSET

$$\{(\mathcal{S}, w) \in \Omega^{m+1} : w \in T^L(\mathcal{S})\} \quad (1)$$

IS A BOREL MEASURABLE SUBSET

VALIDITY QUANTIFIES THE SIZE OF THE SET

$$\{(\mathcal{S}, w) \in \Omega^{m+1} : w \notin T^L(\mathcal{S})\} \quad (2)$$

WHICH IS MEASURABLE BY ASSUMPTION 1

DEFINITION 1

PROBABILITY PRODUCT WHICH
GIVES JOINT DISTRIBUTION
OVER $S \cup \{w\}$ GIVEN $w_1, w_2, \dots, w_m$ IID

FOR ANY DISTRIBUTION $\mathbb{P}^{m+1}$ DEFINED ON Ω^{m+1}
AND ANY **CONFIDENCE LEVEL** $\delta \in (0, 1)$,
A SET PREDICTOR IS SAID TO BE
CONSERVATIVELY VALID AT SIG. LEVEL δ IF:

$$\mathbb{P}^{m+1} \{ (S, w) : w \notin \mathcal{T}^c(S) \} \leq \delta$$

NONCONFORMITY MEASURE

PERMUTATION - INVARIANT, MEASURABLE MAPPING

$\mathcal{A} : \Omega^m \mapsto \mathbb{R}^m$ SUCH THAT,

$\forall \pi$ IN THE PERMUTATION GROUP OF $1:m$

$$\mathcal{A}(w_1, \dots, w_m) = \mathcal{A}(w_{\pi(1)}, \dots, w_{\pi(m)}) \quad (3)$$

PERMUTATION INVARIANCE IS ENFORCED BY **EXCHANGEABILITY**.

$$\text{let } \mathcal{A}(S \cup \{w\}) = [R_1, \dots, R_m, R]^T \in \mathbb{R}^{m+1}$$

WITH $R_i \forall i \in 1:m$ ARE NONCONFORMITY SCORES

ASSUMPTION 1 FOR ANY COLLECTION $S \cup \{w\} \in \Omega^{m+1}$
OF INDEPENDENT SAMPLES FROM $\mathbb{P}$, $R_i \perp R_j \forall i \neq j$
WITH NO REPETITIONS, AND SORTED IN NON-DECREASING ORDER

DEFINE THE FUNCTION:

$$f(w; S) := \frac{|\{i \in 1:(m+1) : R_i > R\}|}{m+1} \quad (4)$$

FRACTION OF THE POINTS
WHOSE CONFIDENCE SCORE IS
HIGHER THAN THAT OF THE TEST
POINT w

WHERE $|\cdot|$ IS THE CARDINALITY FUNCTION, WHICH
COUNTS THE ELEMENTS WITHIN A GIVEN SET

THEN, THE SET PREDICTOR IS CONSTRUCTED AS:

$$T_\delta^c(S) = \{w \in \Omega : f(w; S) > \delta\} \quad (5)$$

WHICH OUTPUTS A RANGE OF POINTS THAT
WILL FAIL TO CONTAIN THE TRUE OUTCOME
WITH A PROBABILITY NO GREATER THAN δ

OR, IN OTHER WORDS, A COLLECTION OF REALIZATIONS
THAT WILL CONTAIN THE TRUE OUTCOME WITH
A PROBABILITY OF AT LEAST $1 - \delta$

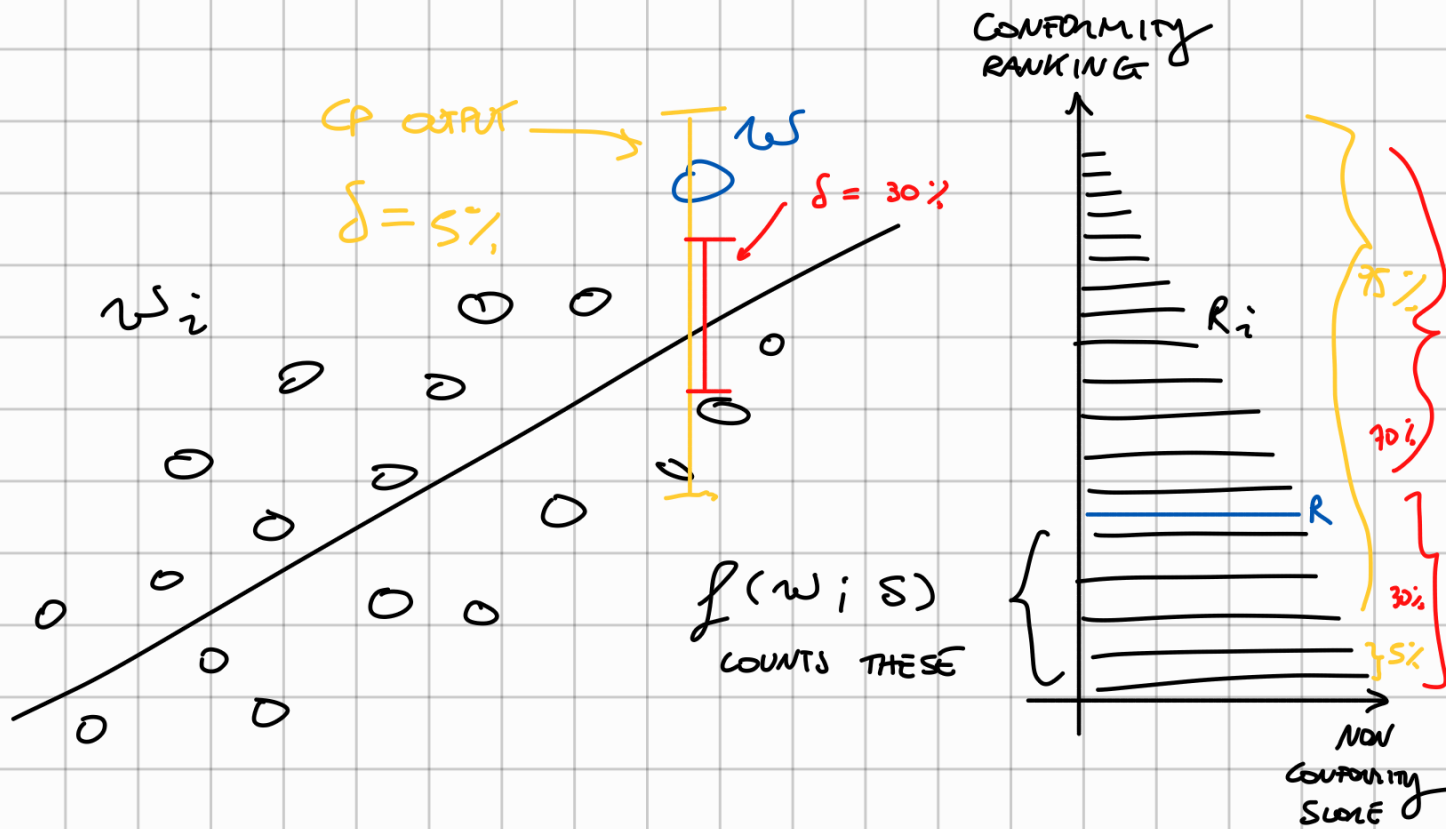
↓ THESE TWO STATEMENTS ARE THEOREM 1

THEOREM 1 (ASSUMING ASSUMPTION 1 HOLDS)

THE SET PREDICTOR T_δ^c IS **CONSERVATIVELY
VALID** AT THE SIGNIFICANCE LEVEL δ

$$\mathbb{P}^{m+1} \{(S, w) \in \Omega^{m+1} : w \notin T_\delta^c(S)\} \leq \delta \quad (6)$$

EXAMPLE: LINEAR REGRESSION



$$f(w; S) = \frac{5}{20} = 0.25$$

E.G. IF $\delta = 0.05$ (5% error rate)

THEN WE WOULD ACCEPT THIS w SINCE IT LIES WITHIN THE 95%. AND

$$f(w; S) > \delta$$

BUT IF WE HAD $\delta = 0.3$ THEN WE WOULD REJECT THIS w

CONSEQUENCE OF THEOREM 1: ^{HOW MANY POINTS HAVE NONCONFORMITY SCORE HIGHER THAN THAT OF THE TEST SAMPLE?}

E.G. $\delta = 0.1 \rightarrow 32$ POINTS
 $m+1 = 325$

(7)

$$\{w \notin T_S^c(S)\} = \left\{w \in \Omega : \left| \{i \in 1:(m+1) : R_i \geq R\} \right| \leq \lfloor \delta(m+1) \rfloor \right\}$$

IN ENGLISH, THE SET OF REALISATIONS WHICH DO NOT BELONG TO THE PREDICTED SET IS EQUAL TO THE SET OF REALISATIONS WITH BOUNDS SUCH THAT THE NUMBER OF POINTS WITH A NONCONFORMITY SCORE HIGHER THAN THAT OF THE TEST POINT IS AT MOST A NUMBER EQUAL TO AN INTEGER PROPORTION δ OF THE $m+1$ BAG OF SAMPLES

LETTING $p = \lceil (1-\delta)(m+1) \rceil$

$$\mathcal{Q} := \{w \in \Omega : R > R_p\} \quad (8)$$

REPRESENTS THE SET OF REALISATIONS WITH A NONCONFORMITY SCORE HIGHER THAN AN INTEGER PROPORTION $1-\delta$ OF THE BAG OF SAMPLES

$$R_p := \text{QUANTILE}_{1-\delta}(R_1, \dots, R_m, \infty)$$

^{CONNECTION TO OBTAIN FINITE SAMPLE FUNCTION}
^{"INFIMUM" or "MINIMUM"}

WITH $\text{QUANTILE}_{1-\delta}(Z_i) := \inf_h \{z \in \mathbb{R} : \mathbb{P}(Z \leq z) \geq 1-\delta\}$

let $Z_0 \sim \sum_{i=0}^m \frac{1}{m+1} \delta_{R_i}$ ^{DIRAC DISTRIBUTION} $\Rightarrow R_p = \text{QUANTILE}_{1-\delta}(Z_0)$

WHICH IMPLIES THERE EXIST $\lceil (1-\delta)(m+1) \rceil$ NONCONFORMITY SCORES LOWER THAN OR EQUAL TO R_p (HENCE OF R) ^{AND}

HENCE, THE NUMBER OF NONCONFORMITY SCORES SUCH THAT $R_i \geq R$ WOULD BE AT MOST

$$m+1 - \lceil (1-\delta)(m+1) \rceil = \lfloor \delta(m+1) \rfloor$$

THIS IS EQUIVALENT TO SAYING

$$|\{i \in 1:(m+1) : R_i \geq R\}| \leq \lfloor \delta(m+1) \rfloor \quad (9)$$

By (7), ANY SUCH w IS NOT IN THE PREDICTION SET, WHICH PROVES:

COROLLARY 1

$$\text{let } p = \lceil (1-\delta)(m+1) \rceil$$

$$\mathbb{P}^{m+1} \{ (S, w) \in \Omega^{m+1} : R > R_p \} \leq \delta \quad (10)$$

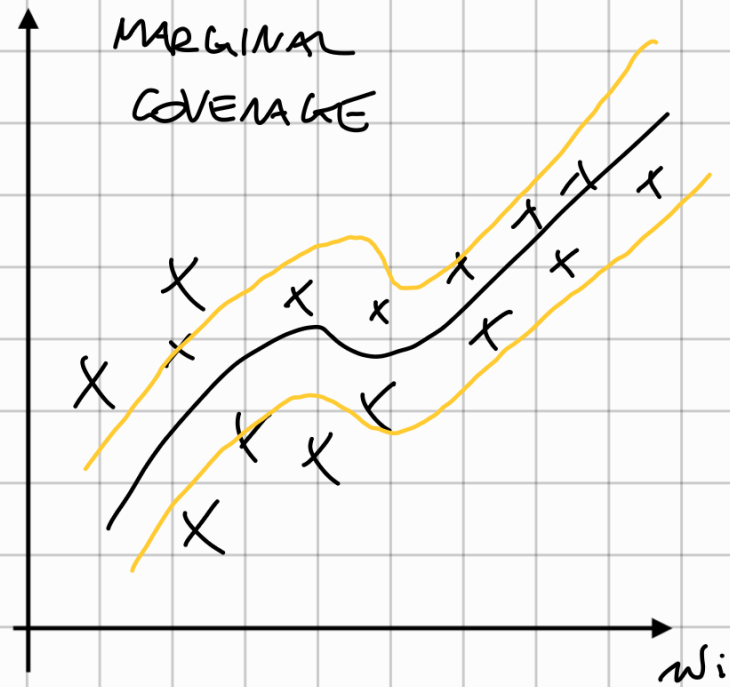
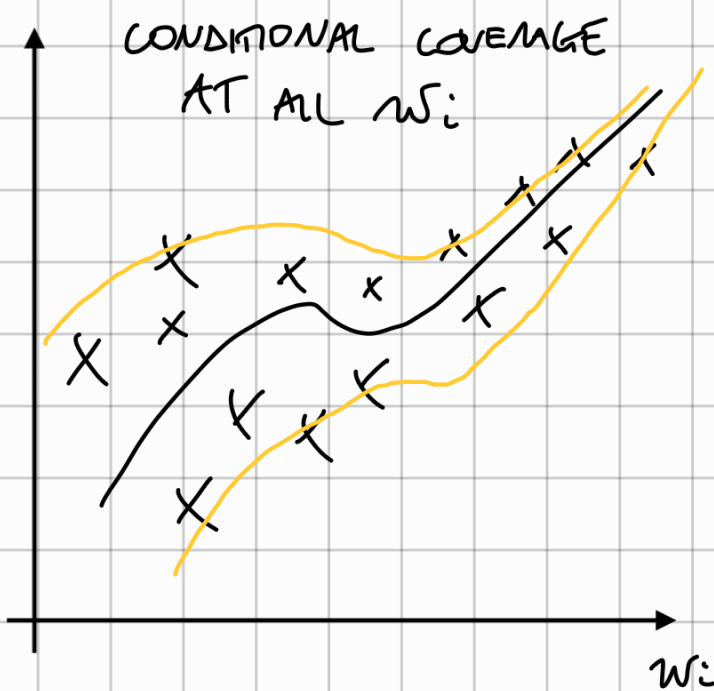
HENCE, THE BOUNDS TO THE PREDICTION SET $\mathcal{T}(S)$ CAN BE EFFICIENTLY COMPUTED AS THE POINT PREDICTION PLUS OR MINUS THE p -TH QUANTILE

IN A WAY, CONFORMAL PREDICTION IS ENGINEERED IN SUCH A MANNER THAT ENSURES CONSERVATIVE VALIDITY HOLDS FOR THE PREDICTION SET

CONDITIONAL VS MARGINAL COVERAGE

SO FAR, ASSESSING CONSERVATIVE VALIDITY HAS ESTABLISHED MARGINAL, BUT NOT CONDITIONAL, COVERAGE GUARANTEES

E.G. REGRESSION POINT PREDICTOR CP - WRAPPER



TO ESTABLISH CONDITIONAL COVERAGE (ON EITHER THE CALIBRATION SET, TEST OBJECT, OR THE TEST NONCONFORMITY SCORES THEMSELVES), WE MUST AIM TO QUANTIFY:

$$\mathbb{P}^{m+1} \left\{ (S, w) \in \Omega^{m+1} : R > R_p \mid S, w, R_1, \dots, R_m, R \right\}$$

IN VOVK 2012 "CONDITIONAL VALIDITY..."

A FULL DESCRIPTION IS MADE, WHERE

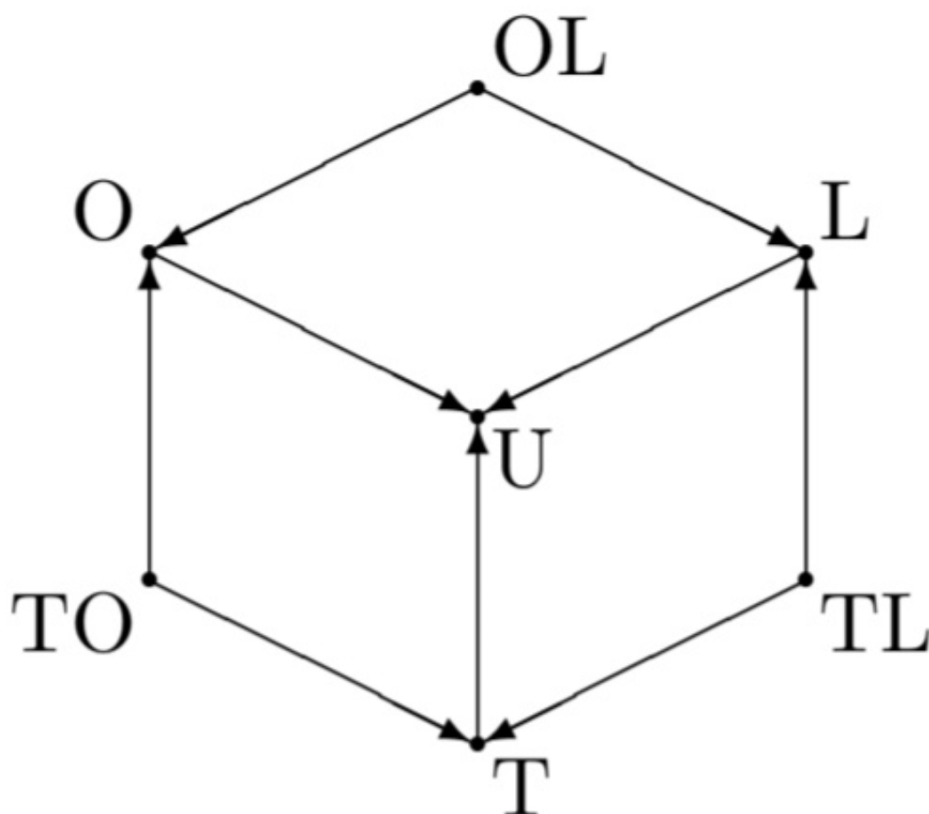
DIFFERENT CONDITIONING PARAMETERS ARE

INDIVIDUALLY CONSIDERED: THE TRAINING SET

T ($s_{a|i}$, IN OUR CASE), THE TEST OBJECT

O (n_5 IN OUR CASE), AND THE

TEST LABEL L (R IN OUR CASE)



CALIBRATION CONDITIONAL CONFORMAL PREDICTION

CALIBRATION CONDITIONAL CP IS AN EXTENSION OF CP TO SOLVE PROBLEMS OF THE FORM:

$$\mathbb{P}_S \{ TP \{ w \in T^L(S) \} \geq 1 - \varepsilon \} \geq 1 - \beta$$

AS : $\nwarrow$ PROBABLY APPROXIMATELY CORRECT FORM

$$R_p := \text{QUANTILE}_{1 - \varepsilon + \sqrt{\frac{\ln(1/\beta)}{2m}}}(R_1, \dots, R_m, \infty)$$

IN PAPER, THIS CORRESPONDS TO:

$$\mathbb{P}_S^m \left[\mathbb{P}_w \{ w \in T^L(S) \} \leq \varepsilon \right] \geq 1 - \delta$$

RISK $\swarrow$
CONFIDENCE $\swarrow$